

# Sergio A. García Mejía

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## EDUCATION

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|-----------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>University of Maryland, College Park</b>                                                                           | Aug. 2025 (expected)           |
| Ph.D., Civil and Environmental Engineering                                                                            | Maryland, USA                  |
| <ul style="list-style-type: none"><li>Graduate Certificate in Latin American &amp; Caribbean Studies (2024)</li></ul> |                                |
| M.Sc., Civil and Environmental Engineering, Water Resources & Disaster Resilience Specialization                      | May 2020                       |
| <ul style="list-style-type: none"><li>Fulbright Foreign Student Program Scholar (2018-2020)</li></ul>                 | Maryland, USA                  |
| <b>Universidad de San Carlos de Guatemala</b>                                                                         | Aug. 2016                      |
| B.S., Civil Engineering (major) and Business Administration (minor)                                                   | Ciudad de Guatemala, Guatemala |

## FELLOWSHIPS AND AWARDS

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|----------------------------------------------------------------------------------------|------------------------|
| <b>NOAA Sea Grant Knauss Executive Fellow</b> , Bureau of Ocean Energy Management      | Jan. 2022 – Jan. 2023  |
| <b>Fulbright Foreign Student Program Scholar</b> , University of Maryland              | July 2018 – July 2020  |
| Graduate Resident, Latin American and Caribbean Studies Center, University of Maryland | Jan. 2020 – July. 2020 |
| Weather Research Instrument Research and Data Publication, Natural Hazards Center      | June 2021              |
| LACS Pop-Up Research Grant, Latin American and Caribbean Students Center, UMD          | May 2023               |
| Jacob K. Goldhaber Travel Grant, The Graduate School, UMD                              | May 2023               |
| International Conference Student Support Award (ICSSA), The Graduate School, UMD       | June 2023              |

## RESEARCH & TECHNICAL EXPERIENCE

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| <b>University of Maryland, Department of Civil and Environmental Engineering (CEE)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Sep. 2020 – Present   |
| <i>Graduate Research Assistant, Ph.D.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Maryland, USA         |
| <ul style="list-style-type: none"><li>Conducted a detailed literary review on the intersections of pro bono work and engineering.</li><li>Contributed to the design of a non-probabilistic survey that was distributed nationally (in the U.S.) to study the perspectives of civil engineers on pro bono work using Qualtrics.</li><li>Created databases of disaster impacts using information from digital newspapers (via Google Boolean searches) from 2015 to 2021 in Python, R, and Office.</li><li>Led two teams of Spanish-speaking and English-speaking interpreters to conduct an in-depth textual analysis of over 2,000 emergency communication messages during the Eta and Iota storms, generating insights from the analysis, organizational reports, documents, and literature reviews using Office, R, Qualtrics, and Dedoose. Mentored 2 undergraduate researchers.</li><li>Mentored 3 undergraduate senior students from CEE and 1 undergraduate student on her 2021 RISC Summer fellowship in quantitative research using Python, ArcGIS, and how to perform literature reviews.</li><li>Facilitated introductions to clean energy and power generation to high school students, fostering interest in clean and sustainable energy careers and supporting workforce diversity in the industry.</li><li>Led a team in designing a website and developing 4 educational modules on nuclear energy for K-12, high school, and college programs under the Supporting Strategic Training of Adaptable and Integrated Nuclear Workforce program for the renewable energy industry.</li></ul> |                       |
| <i>Graduate Research Assistant, M.Sc.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Jan. 2019 – Aug. 2020 |
| <ul style="list-style-type: none"><li>Collected, digitized, and translated data from population and housing censuses of the last 50 years in Guatemala</li><li>Analyzed the impacts of disasters in Guatemala through statistical (Matlab) and spatial (ArcGIS) analysis methods intersecting multiple social, demographic, infrastructure, and housing characteristics variables.</li><li>Undertook a literature analysis on the interplay of housing components and disaster impacts.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                       |
| <b>Structural Extreme Events Reconnaissance (StEER) Network</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Sep. 2021 – Present   |
| <i>Member / Virtual and Field Assessments Contributor</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Remote / Global       |
| <ul style="list-style-type: none"><li>Worked with virtual response teams of Nippes Earthquake (Haiti, 2021), Hualien City Earthquake (Taiwan, 2024), and Hurricane Helene (USA, 2024).</li><li>Contributed to 2 media repositories and 1 preliminary virtual reconnaissance report.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                       |
| <b>Environmental Justice Policy Analyst</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Feb. 2022 – Feb. 2023 |
| <i>John A. Knauss Fellowship, Bureau of Ocean Energy Management (BOEM)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Washington D.C., USA  |

- Led workshops, discussions, and training sessions on methodological processes for quantitative and qualitative analysis of Environmental Justice criteria in BOEM offshore wind energy projects.
- Conducted annotated bibliographies and wrote a report on best practices for Environmental Justice analysis in National Environmental Policy Act BOEM projects.
- Represented BOEM in multiple interagency committee meetings, such as the White House Environmental Justice Advisory Council (WHEJAC), the Natural Disasters & EJ Subcommittee, the USGS Risk Community of Practice, the Science for Disaster Risk Reduction (SDR) Interagency Working Group and the Justice40 Subcommittee.
- Represented BOEM at the Alaska Federation of Natives Convention (Anchorage, AK, 2022).

## PUBLICATIONS

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### MSc Thesis

- 2021 **García Mejía, Sergio**. Building Resilience or Building Fragility? Understanding Disaster Resilience Patterns in Guatemala through the Analysis of Disaster Datasets in Connection with Population and Housing Census Information. University of Maryland. <https://doi.org/10.17603/ds2-29ee-1862>

### Dataset and instrument

- 2021 **García Mejía, S.**, N. Ghosh, M. Bensi. Leveraging digital news to create databases of the impacts of small, medium and, large disasters. DesignSafe-CI. <https://doi.org/10.17603/ds2-29ee-1862>

### Reports

- 2024 Kyprioti, A., J. Yuzbasi, H. Dang, A. Jana, S. Kameshwar, A. Diekmann, **S. Garcia**, [full list of authors in the link]. "StEER: Hurricane Helene Annotated Media Repository", in StEER - Hurricane Helene. DesignSafe-CI. <https://doi.org/10.17603/ds2-mxeh-7712>
- 2024 Wolohan, K., K. Mosalam, A. Jana, N. BEKTAS, K. Erazo, **S. Garcia**, [full list of authors in the link]. "StEER: 2024 Hualien City Earthquake Annotated Media Repository", in StEER- 2024 Hualien City, Taiwan Earthquake. DesignSafe-CI. <https://doi.org/10.17603/ds2-4xv5-qc41>
- 2023 Safiey, A., D. Roueche, L. Huallpa, W. Hassan, C. Reis, S. Sorosh, M. Lopez Ruiz, Y. Merino-Peña, L. Vargas, A. Dilsiz, A. Martin, J. Cohen, Z. Khalil, B. Tunc, **S. Garcia**, [full list of authors in the link]. "StEER: 2023 Mw 7.8 Kahramanmaras, Turkiye Earthquake Sequence Preliminary Virtual Reconnaissance Report (PVRR) Media Repository", in StEER- February 6, 2023, Kahramanmaras, Türkiye, Mw 7.8 Earthquake [Version 2]. DesignSafe-CI. <https://doi.org/10.17603/ds2-mm04-xq43>
- 2021 Kijewski-Correa, T., B. Alhawamdeh, C. Arteta, W. DJIMA, T. Do, **S. Garcia**, [full list of authors in the link]. "StEER: M7.2 Nippes, Haiti Earthquake Preliminary Virtual Reconnaissance Report (PVRR)", in StEER - 14 August 2021, M7.2 Nippes Earthquake, Haiti. DesignSafe-CI. <https://doi.org/10.17603/h7vg-5691>

### In Preparation

- **García Mejía, S.**, Herovic, E., Bensi, M. Short Messaging Emergency Communication Practices during Tropical Cyclones Eta and Iota in Guatemala. Target journal: International Journal of Disaster Risk Reduction. Research paper to be submitted by August 22nd, 2025.
- **García Mejía, S.**, Ault, A., Bensi, M. Reducing Information Gaps in Critical Infrastructure Through Geocoding and Surrogate Addressing Protocols. Target journal: International Journal of Disaster Risk Reduction. Research paper to be submitted by September 19th, 2025.
- **García Mejía, S.**, Ghosh, N., Bensi, M. Leveraging Digital News to Create Databases of the Impacts of Small, Medium, and Large Disasters. Target journal: Computers, Environment and Urban Systems. Research paper to be submitted in November 2025.
- **García Mejía, S.**, Marandi, S., Walsh, M., Bensi, M., Modarres, M. "Addressing Data Gaps in U.S. Nuclear Fuel Workforce and Production: Collection Challenges and Implications for Energy Planning". Conference paper to be submitted by June 24, 2025, for the ANS 2025 International Conference on Accelerator Applications (AccApp'25).

## WATER RESOURCES / CIVIL & ENVIRONMENTAL ENGINEERING EXPERIENCE

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### **Municipal Government of Jocotenango**

Jan. 2017 – July 2018

#### *Planning Department Supervisor*

Jocotenango, Sacatepequez, Guatemala

- Supervised the construction of seven different municipal projects for a total value of \$1M.

- Led the design of sewage water treatment plants, water supply systems, stormwater management, riverbank protection structures, and sports field drainage systems. Used AutoCAD and Office tools.
- Led the financial planning of municipal projects for the year 2019 through intra-institutional coordination and the ministries of education, health, culture, and sports. Used Microsoft Excel for budgeting.
- Trained 6 teammates from the Department of Municipal Planning to implement standardized project design processes using AutoCAD, Excel, a systematized collective design protocols.

**Agua-Info, S.A.,** 26 Ave. 10-76 zona 7 Kaminal Juyu 1, Ciudad de Guatemala

Aug. 2016 – Dec. 2016

*Field Researcher*

Ciudad de Guatemala, Guatemala

- Measured and analyzed the hydraulic inventory (open channels, pipelines) of sugarcane plantations in Guatemala.
- Compiled the final report detailing the results of the analysis using H&H models and GIS spatial analysis.

**Ing. Fredy García Fuentes,** Vía Bilbao #1 Col. Las Terrazas, Cdad. San Cristobal, Mixco, Guatemala

Feb. 2013 – Dec. 2014

*Junior Civil Engineer - Supervisor*

Mixco, Guatemala

- Supervised the maintenance of the principal and secondary roadways in the Municipalities of Sacatepéquez and San Marcos.
- Oversaw the construction of gabion walls for riverbank protection in Tiquisate and Chiquimula.

**Municipality Government of Santa Catarina Pinula**

Jul. 2012 – Feb. 2013

*Junior Civil Engineer*

Santa Catarina Pinula, Guatemala

- Collaborated on the design of municipal projects, including water supply systems for rural populations, small-scale sewage water treatment plants, schools, hospitals, parks, and roads

## CONFERENCE PRESENTATIONS

|                                                                                                                                                                    |                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| <b>Natural Hazards Center Researchers' Workshop,</b> Researchers Meeting                                                                                           | July 2021                  |
| • <i>"Emergency Disaster Shelter Use in Guatemala (2012-2018): A Protocol for Web-Scraping Geolocations"</i>                                                       |                            |
| <b>Natural Hazards Center Researchers' Workshop,</b> Researchers Meeting                                                                                           | July 2022                  |
| • <i>"Understanding Emergency Communication Practices During Hurricanes Eta and Iota in Guatemala"</i>                                                             |                            |
| <b>Society for Risk Analysis Annual Meeting,</b> Tampa Bay, FL                                                                                                     | Dec. 2022                  |
| • <i>"Emergency Communication Strategies During 'Back to Back' Tropical Cyclones Eta and Iota"</i>                                                                 |                            |
| <b>Latin American Studies Association,</b> Annual Congress, Vancouver, Canada                                                                                      | May 2023                   |
| • <i>"Communicating through Consecutive Crises in Guatemala: Tropical Storms Eta and Iota"</i>                                                                     |                            |
| <b>Latin American Social and Public Policy Conference,</b> University of Pittsburgh                                                                                | Mar. 2024                  |
| • <i>"Construir ¿Resiliencia o fragilidad? Patrones de resiliencia ante los desastres en Guatemala" (Spanish)</i>                                                  |                            |
| <b>Central American Politics Consortium,</b> Tulane University                                                                                                     | Jan. 2024                  |
| • <i>"Environmental justice considerations for the environmental impact analysis instruments in Guatemala" (Spanish)</i>                                           |                            |
| <b>ANS/ASME Joint Committee on Nuclear Risk Management Meeting,</b> University of Maryland                                                                         | Feb. 2025                  |
| • <i>"Developing Educational Content on Nuclear Energy to Engage a Diverse Workforce"</i>                                                                          |                            |
| <b>Latin American Studies Association,</b> Annual Congress, San Francisco, CA                                                                                      | May 2025                   |
| • <i>"Temporal variability of environmental flows in connection to policy-making and hydropower licensing in top-down hydrocracies: A case study in Guatemala"</i> |                            |
| <b>American Nuclear Society,</b> 2025 ANS Winter Conference and Expo, Washington D.C.                                                                              | (in preparation) Nov. 2025 |
| • <i>"Addressing Data Gaps in U.S. Nuclear Fuel Workforce and Production: Collection Challenges and Implications for Energy Planning"</i>                          |                            |
| <b>Society for Risk Analysis Annual Meeting,</b> Washington D.C.                                                                                                   | (in preparation) Dec. 2025 |
| • <i>"Refining Short-Alert Best Practices: A Spanish-English Comparison of Tropical-Cyclone Short Emergency Communication Messages"</i>                            |                            |

## CONFERENCE POSTERS

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|-------------------------------------------------------------------------------------------------------|------------|
| <b>Natural Hazards Research Summit,</b> Natural Hazards Engineering Research Infrastructure (NHERI)   | Oct. 2022  |
| <b>American Society of Civil Engineers INSPIRE,</b> 2023 Conference                                   | Nov. 2023  |
| <b>Annual Symposium from the Department of Civil and Environmental Engineering,</b> UMD, College Park | April 2024 |
| <b>Natural Hazards Research Summit,</b> Natural Hazards Engineering Research Infrastructure (NHERI)   | May 2024   |

## INVITED TALKS AND WORKSHOPS

|                                                                                                                                                                                                         |           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>Latin American &amp; Caribbean Studies Center</b> , University of Maryland                                                                                                                           | Oct. 2021 |
| <ul style="list-style-type: none"> <li>Central American Week panel. “Understanding the impacts of consecutive disasters and the use of emergency shelters”</li> </ul>                                   |           |
| <b>Latin American &amp; Caribbean Studies Center</b> , University of Maryland                                                                                                                           | Mar. 2022 |
| <ul style="list-style-type: none"> <li>World Water Day panel. “Disproportionate and highly adverse impacts on vulnerable populations during 2020 back-to-back storms in Guatemala”</li> </ul>           |           |
| <b>Latin American &amp; Caribbean Studies Center</b> , University of Maryland                                                                                                                           | Mar. 2022 |
| <ul style="list-style-type: none"> <li>Graduate Lecture Series. “Disaster Risk Reduction in Guatemala: Case Studies on Information, Communication, and Memory Systems”</li> </ul>                       |           |
| <b>Centro de Estudios Urbanos y Regionales (CEUR)</b> , Universidad de San Carlos de Guatemala                                                                                                          | May 2021  |
| <ul style="list-style-type: none"> <li>Provided commentary on Jorge Aragón’s newest book, “Territorio y Metrópolis en Guatemala y Centroamérica”</li> </ul>                                             |           |
| <b>Centro de Estudios Urbanos y Regionales (CEUR)</b> , Universidad de San Carlos de Guatemala                                                                                                          | Nov 2022  |
| <ul style="list-style-type: none"> <li>Provided commentary on Amanda Moran &amp; Jorge Aragón’s newest book, “Movilidad Urbana Sostenible para el Área Metropolitana de Ciudad de Guatemala”</li> </ul> |           |
| <b>Instituto de Investigación en Ciencias Socio Humanistas</b> , Universidad Rafael Landívar                                                                                                            | Jan. 2024 |
| <ul style="list-style-type: none"> <li>Invited talk. “¿Resiliencia o Fragilidad? Impacto social de desastres en Guatemala y patrones de respuesta”</li> </ul>                                           |           |

## SKILLS

**Software:** R, Matlab, Python, HEC-RAS, HEC-HMS, EPANET, SWMM, HY8, ArcGIS, QGIS, AutoCAD Civil3D, SPSS/STATA, @Risk, DecisionTools, Photoshop, Premiere Pro/Express, Qualtrics, Office, Lucidchart, Canva, WordPress, Github, Gitlab.  
**Language:** English (Proficient), Spanish (Native), Portuguese (Intermediate), Kaqchikel (Beginner).

## ACADEMIC SERVICE

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| <b>Latin American Studies Center (LACS)</b> , University of Maryland, College Park                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Aug. 2018 - Present      |
| <i>Graduate Student Committee Member / Mentor</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Maryland, USA            |
| <ul style="list-style-type: none"> <li>Co-organized 4 student conferences between 2019 and 2023 as a member of the organizing committee, along with several graduate student multidisciplinary series talks.</li> <li>Mentored 3 undergraduate students on their research capstone projects for their certificate program.</li> <li>Co-wrote the LACS-internal white paper “Multilevel Research Mentorship in the Latin American and Caribbean Center at the University of Maryland, College Park.” (June 2022)</li> </ul>                                                                                                                                                                                                                                  |                          |
| <b>Department of Civil &amp; Environmental Engineering (CEE)</b> , University of Maryland, College Park                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Aug. 2022 - Present      |
| <i>Graduate Student Council Member</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Maryland, USA            |
| <ul style="list-style-type: none"> <li>Co-led initiatives to strengthen graduate student community engagement by organizing multiple social events and networking opportunities.</li> <li>Collaborated in organizing departmental meetings focused on improving services, resources, and professional development opportunities for graduate students in Civil and Environmental Engineering.</li> </ul>                                                                                                                                                                                                                                                                                                                                                    |                          |
| <b>Latin American Studies Association (LASA)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Aug. 2023 – May 2025     |
| <i>LASA Central America Section (CAS-LASA) Board Member, Student Representative</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Pittsburgh, USA / Remote |
| <ul style="list-style-type: none"> <li>Co-directed scholarly exchange within CAS-LASA’s 200+ members, strengthening networks in Central American studies.</li> <li>Co-organized the sponsorship of 3 panels for LASA2024 (Bogotá, Colombia) and 3 panels for LASA2025 (San Francisco, CA, USA), including the call for papers, selection, and communications through the CAS-LASA listserv and social media.</li> <li>Co-led 19 monthly newsletters to promote academic exchange and strengthen CAS-LASA’s digital community.</li> <li>Coordinated \$4,200 in equitable travel grants for 6 awardees by co-leading an inclusive call, curating proposals, and collaborating with institutional partners to ensure transparent fund disbursement.</li> </ul> |                          |